Technical Document #326: Cloud Computing - Comprehensive Overview

Technical Document #326: Cloud Computing - Comprehensive Overview

Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet. AWS EC2 and Azure VMs are popular IaaS offerings.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Auto-scaling automatically adjusts compute resources based on demand.
- Serverless computing allows developers to run code without managing servers.
- Kubernetes orchestrates containerized applications across clusters of machines.